# 🌐 VLANs Part 3 (10.31.2026)

## 📋 Video Overview

This video covers advanced VLAN topics for CCNA, including:

- Native VLAN configuration on routers (ROAS)
- Wireshark captures showing 802.1Q tags
- Layer 3 switching (Multilayer switching) for inter-VLAN routing
- Introduction to SVIs (Switch Virtual Interfaces)

> 📝 **Note:** DTP and VTP are reserved for a separate video (removed from CCNA exam topics but still useful to understand).

## 🏷️ Native VLAN on a Router (ROAS)

### What is the Native VLAN benefit?

- Frames in the native VLAN are **untagged** → more efficient
- Smaller frames = more frames per second

## ⚠️ Security Best Practice

> Set native VLAN to an **unused VLAN** for security purposes.

### Two Methods to Configure Native VLAN on Router

#### Method 1: Using Subinterface with `NATIVE` Keyword

```
interface g0/0.10
 encapsulation dot1q 10 native
```

#### Method 2: Configure IP on Physical Interface

```
interface g0/0
 ip address 192.168.1.1 255.255.255.192
```

- No subinterface needed
- No `encapsulation dot1q` command required

## 🔍 Wireshark Capture Analysis

### Tagged Frame (VLAN 20 - NOT native)

| Field | Value | Meaning |
|---|---|---|
| **TPID** | 8100 | Indicates 802.1Q tag |
| **PCP** | 0 | No special priority |
| **DEI** | 0 | Won't be dropped during congestion |
| **VLAN ID** | 20 | The VLAN this frame belongs to |

### Untagged Frame (VLAN 10 - Native VLAN)

- No 802.1Q tag present
- Both router and switch understand untagged frames = VLAN 10
- More efficient transmission

## 📊 Frame Flow Example

1. PC in VLAN 20 pings PC in VLAN 10
2. ICMP request: **tagged VLAN 20** (SW2 → R1)
3. ICMP request: **untagged** (R1 → SW2, native VLAN 10)
4. Continues untagged to destination
5. Reply follows reverse path

# 🔄 Layer 3 Switching (Multilayer Switching)

## What is a Multilayer Switch?

- Capable of **both switching AND routing**
- **Layer 3 aware** (unlike regular Layer 2 switches)
- Can assign IP addresses to interfaces
- Can configure **routed ports**
- Can create **virtual interfaces for each VLAN** (SVIs)
- Supports static routes
- Used for **inter-VLAN routing**

## Inter-VLAN Routing Methods Comparison

| Method | Pros | Cons |
|---|---|---|
| **Multiple connections** | Simple | Limited by router interfaces |
| **Router on a Stick (ROAS)** | Efficient interface usage | Network congestion possible |
| **Layer 3 Switch (MLS)** | Best for large networks | More complex configuration |

## 📊 Traffic Path Comparison

**Router on a Stick:**

```
PC (VLAN20) → SW2 → R1 (tagged VLAN20) → SW2 (tagged VLAN10) → SW1 → PC (VLAN10)
```

**Layer 3 Switch:**

```
PC (VLAN20) → SW2 (routes internally) → SW1 (tagged VLAN10) → PC (VLAN10)
```

> ✅ Traffic never goes to R1 for inter-VLAN routing!

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# ⚙️ Layer 3 Switch Configuration

## Step 1: Configure Point-to-Point Link (SW2 ↔ R1)

**On R1:**

```
no interface g0/0.10
no interface g0/0.20
no interface g0/0.30
default interface g0/0
interface g0/0
 ip address 192.168.1.194 255.255.255.252
```

**On SW2:**

```
default interface g0/1
ip routing                          ← CRITICAL! Enables Layer 3 routing
interface g0/1
 no switchport                      ← Converts to routed port
 ip address 192.168.1.193 255.255.255.252
```

> ⚠️ **Important:** Don't forget `ip routing` or inter-VLAN routing won't work!

## Step 2: Configure Default Route on SW2

```
ip route 0.0.0.0 0.0.0.0 192.168.1.194
```

## Step 3: Configure SVIs (Switch Virtual Interfaces)

```
interface vlan10
 ip address 192.168.1.62 255.255.255.192
 no shutdown

interface vlan20
 ip address 192.168.1.126 255.255.255.192
 no shutdown
```

```
interface vlan30
 ip address 192.168.1.190 255.255.255.192
 no shutdown
```

> 📝 **Remember:** SVIs are **shutdown by default** — use `no shutdown`!

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## 📜 SVI Conditions for UP/UP Status

An SVI requires **ALL** of the following to be UP/UP:

| # | Condition | Notes |
|---|---|---|
| 1️⃣ | **VLAN must exist** on the switch | Switch won't auto-create VLAN for SVI |
| 2️⃣ | **At least one access port** in the VLAN is up/up, OR a **trunk port** allowing that VLAN is up/up | |
| 3️⃣ | **VLAN must not be shutdown** | Different from SVI shutdown |
| 4️⃣ | **SVI must not be shutdown** | Use `no shutdown` |

### ⚠️ Important Distinction

- Assigning an access port to a VLAN **auto-creates** the VLAN
- Creating an SVI for a non-existent VLAN **does NOT** auto-create the VLAN

### Verification Commands

```
show ip interface brief
show interfaces status          ← Look for "routed" in VLAN column
show ip route                   ← Shows connected/local routes for SVIs
```

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## 🔑 Key Commands Summary

| Command | Purpose |
|---|---|
| `encapsulation dot1q [vlan-id] native` | Configure native VLAN on subinterface |
| `ip routing` | Enable Layer 3 routing on switch |
| `no switchport` | Convert interface to routed port |
| `interface vlan [id]` | Create/configure SVI |
| `no shutdown` | Enable SVI |
| `ip route 0.0.0.0 0.0.0.0 [next-hop]` | Configure default route |

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## 📝 Quiz Questions

### Question 1

**Which TWO answers are valid options to configure the native VLAN on a router in a ROAS configuration? (select two, each answer is a complete solution)**

Options:

- A) Configure `encapsulation dot1q 10` on subinterface
- B) Configure `encapsulation dot1q 10 native` on subinterface
- C) Configure IP address on physical interface (no subinterface)
- D) Configure `switchport trunk native vlan 10` on router

**B and C**

- **B:** Uses `encapsulation dot1q 10 native` on the subinterface
- **C:** Configures IP address on the physical interface (no encapsulation command needed)

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### Question 2

**You create an SVI for VLAN225 on SW1, assign an IP address, and enable it with no shutdown, but the interface remains down/down. Which TWO options might be causing this? (select two)**

Options:

- A) VLAN225 doesn't exist on the switch
- B) You didn't issue `switchport mode trunk` on VLAN225's SVI
- C) You didn't issue `switchport access vlan 225` on VLAN225's SVI

- D) No interfaces in VLAN225 are up/up

**A and D**

- **A:** The VLAN must exist on the switch for the SVI to come up
- **D:** There must be at least one access port or trunk port in that VLAN that is up/up

> ❌ B and C are incorrect — you don't configure switchport commands on SVIs.

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## Question 3

**Which command is used to configure a switch interface as a routed port?**

Options:

- A) `no switchport`
- B) `ip address [address] [mask]`
- C) `ip routing`
- D) `switchport mode route`

**A)** `no switchport`

This command converts the interface from a Layer 2 switchport to a Layer 3 routed port, allowing you to configure an IP address.

> ❌ C (`ip routing`) enables routing globally on the switch but doesn't set individual interfaces as routed ports.

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## 🎯 Bonus Question (Boson ExSim)

**You issue the following commands on a Catalyst 2950 switch:**

```
configure terminal
interface FastEthernet 0/7
switchport trunk encapsulation dot1q
switchport mode trunk
switchport trunk native vlan 44
```

**Which statement is true regarding VLAN traffic when sent over port FastEthernet0/7?**

Options:

- A) VLAN 1 traffic will be untagged
- B) VLAN 44 traffic will be untagged
- C) All VLAN traffic will be tagged
- D) All VLAN traffic will be untagged

**B) VLAN 44 traffic will be untagged**

- The command `switchport trunk native vlan 44` changes the native VLAN from the default (VLAN 1) to VLAN 44
- Traffic in the native VLAN is **never tagged** when sent over a trunk
- By default, VLAN 1 would be untagged, but the configuration changed it to VLAN 44

> 💡 **Verification:** Use `show interfaces trunk` to see native VLAN settings.

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## 📌 Quick Review Summary

| Topic | Key Takeaway |
|---|---|
| **Native VLAN on Router** | Two methods: subinterface with `native` keyword OR IP on physical interface |
| **Wireshark 802.1Q** | TPID (8100), PCP, DEI, VLAN ID fields visible in tagged frames |
| **Layer 3 Switch** | Can route between VLANs using SVIs — more efficient than ROAS |
| **SVI Requirements** | VLAN exists + port up in VLAN + VLAN not shutdown + SVI not shutdown |
| **Critical Commands** | `ip routing` (global), `no switchport` (interface), `no shutdown` (SVI) |